

School of Science and Technology

COURSEWORK ASSESSMENT SPECIFICATION

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Module Code	COMP30271
Module Title	Cognitive Computing
Module Leader	Dr. João Filipe Ferreira
Module Team	Dr. João Filipe Ferreira
Coursework Title	Portfolio Project
Module Learning Outcomes assessed	MLO1, Systematically and critically understand the concept of cognitive systems and develop the intuition on the difficulty in reaching a universal consensus concerning a strict qualification of anything as being “cognitive”., MLO2, Demonstrate awareness of the different main paradigms corresponding to each of the several disciplines involved (computer science, digital electronics, AI, robotics), so as to be able to contextualise cognitive systems research and its current outlook. MLO3, Understand and evaluate the different architectures that represent the foundations of CC & ACS. MLO4, Systematically and critically understand most of the concepts underpinning CC & ACS, namely: autonomy, embodiment, learning, memory, knowledge and its representation, and social cognition. MLO5, Understand and explain the applications areas of CC & ACS, and also the consequent professional, ethical and legal issues., MLO7, Demonstrate the capacity of being able to independently research topics relating to CC & ACS, in particular human-machine interfaces (HMI), robotics and the “Internet of Things” (IoT).,MLO8, Demonstrate in-depth skills in system architecture design, mathematical modelling and programming required to implement CC & ACS., MLO9, Demonstrate the capacity of taking in and integrating know-how both from computer science and engineering and from human sciences and nature.
Apprentice Learner KSBs evidenced	N/A
Contribution to module (include contribution to element if appropriate)	40%
Date work set	27/10/2025
Deadline for submissions	13/02/2026 2.30 pm
Method of submission	Dropbox via NTU NOW
Deadline for feedback	3 weeks following submission deadline
Method of feedback	Formal feedback and grades will be provided via Dropbox comments.
Previous coursework that may support this submission.	N/A

Late submissions and Considerations of Personal Circumstances

Work handed in up to five working days late will be given a maximum grade of Pass whilst work that arrives more than five working days will be given a grade of Zero.

Please note, if you are repeating the work and are capped in your grade, then you will receive a Zero grade if the work is submitted after the initial deadline.

Work will only be accepted beyond the five working day deadline if you have asked for an extension in consideration of your personal circumstances. That means something has occurred, beyond your control, that has impacted your ability to complete the work. Details of the process of requesting extensions and what is considered valid reasons to request can be found here in the [Students' Handbook](#)

Breaches of Academic Integrity

To ensure that you are not accused of any breaches of Academic Integrity, look at the NOW page [Plagiarism and Academic Integrity at NTU](#) for guidance.

The University views **plagiarism and collusion** as serious academic irregularities and there are a number of different penalties which may be applied to such offences. [Section 17C](#) of the Quality Handbook outlines the penalties and states on page 21-22 that **plagiarism** includes:

- “presenting someone else’s ideas as your own [**including text, graphs, diagrams, videos, etc.**] in a substantial proportion of your work, with or without consent, by incorporating it into an assessment without full acknowledgement, including:
- **self-plagiarism**: reproducing or representing work for assessment without proper attribution and attempting to gain credit for this work where credit has already been received;
- **paraphrasing**: rephrasing a source’s ideas without proper attribution;
- **mosaic plagiarism/patchworking**: weaving phrases and text from several sources into your own work; and/or adjusting sentences without quotation marks or attribution;
- **source-based plagiarism**: providing inaccurate or incomplete information about sources such that they cannot be found;
- **computer code plagiarism**: copying or adapting source code without permission from and attribution to the original creator;”

whereas **collusion** includes:

- “working with other students on an assessment meant for individual submission;
- sharing your work with other students enabling them to plagiarise your ideas.”

In particular, please note that submitting portions of work already assessed for the same learning outcomes is **self-plagiarism** and is a serious breach of academic integrity.

Penalties for breaches of academic integrity range from capped or zero grades for elements of modules, to dismissal from the course and termination of studies.

AI-powered language models

At NTU, we use [Microsoft Copilot](#). This is included within the institutional Microsoft Subscription and by logging into Microsoft Copilot using your NTU login credentials, you can be assured that your use of the tool will be more secure than that of other generative AI services that operate outside of the NTU digital ecosystem.

At its heart, Copilot is a chatbot that can assist you with an almost endless range of tasks. If you're new to Copilot, or Generative AI chatbots in general, Microsoft's three introductory videos are a great place to start:

- [Meet Microsoft Copilot](#)
- [What you can do with Copilot](#)
- [Start using Copilot](#)

We use Copilot over ChatGPT etc as NTU has a data security agreement with Microsoft, which ensures enhanced protection when working within the NTU digital ecosystem. You can upload work documents to Copilot but only upload what is strictly necessary.

You may see instances of Copilot integrated directly into Microsoft apps such as Word, PowerPoint and Excel. NTU's licence doesn't currently provide this level of integration. For now, we only have access to the stand-alone Copilot app.

To ensure you are using Copilot within the safety of the NTU ecosystem, please [login to Copilot](#) using your NTU credentials

Other GenAI Tools

While other GenAI tools may be suitable for general purposes, but you must not upload NTU documentation or any other forms of intellectual property to these platforms. This is essential to keeping everyone's data and resources secure.

Using AI Responsibly

General Guidelines:

- **Transparency:** Always be transparent about your use of AI. If in doubt, seek clarification from the author of this assessment and reference use appropriately.
- **Academic Integrity:** AI tools can support your learning, but they must not replace your own critical thinking and analysis. Never use GenAI to produce your submission for you.
- **Ethical Use:** Follow ethical guidelines when using AI, avoiding plagiarism and misrepresentation of AI-generated content as your own.
- **Limitations:** Never trust an output from GenAI without checking it first. They are poor at summarising resources correctly so you would need to check anyway.

The Traffic Light System is a simple framework used at many institutes to clarify when and how you can use generative AI in assessments. It categorises AI use into three levels: Green, Amber and Red

In this assignment:

Green We encourage the use of GenAI to aid with

Checking spelling and grammar but not in designing structure of the reports. There is a template for that. Reference and use of acknowledgement form needed

Amber We allow the use of GenAI to carry out the following tasks, but you must reference and keep outputs and mention in acknowledgement form.

- Research and data gathering (e.g., asking AI to summarise existing studies or generate ideas, but these must be critically evaluated as are often incorrect)
- Drafting essays or reports (e.g., using AI to help structure or rephrase your work, but with clear identification of AI contributions)

Red You must not use GenAI to

- Complete your work entirely without critical input from yourself
- Complete your reflection aspect
- Generate fake resources for the client
- Generate fake data or to analyse data collected

It is important to note when using any AI platform that they generate the most common responses to questions, not necessarily the correct ones. They also fabricate evidence. The material they produce is not your own words. Assessments require you to answer questions giving your own view and in your own words. The outputs from platforms such as ChatGPT do not provide that.

By presenting such material as your own words you are violating the Academic Integrity policy, a matter that NTU takes very seriously.

The skills you develop during your time with us allow you to interrogate material and evaluate it, which are important skills in all careers. Generative AI does not enable you to develop these skills. If you have utilised such platforms, **you must retain all outputs from them** to provide evidence that your work is your own in the case of suspected breaches of Academic Integrity.

Acknowledging use of GenAI

You must also acknowledge if and how you have used GenAI by completing the GenAI usage acknowledgement form and then submit it with your work to the same dropbox as the assignment. There is no penalty to your grade for not submitting the completed form, but if there is a question about inappropriate use of GenAI then lack of completion will be considered as evidence in any investigation.

Where GenAI is permitted, please follow the guide at the below link to correctly reference the use of GenAI as part of your assignment as you would any reference or resource.

[Referencing of GenAI](#) Then click on **Quick start to Harvard referencing** and look at the bottom of the list.

I. Assessment Requirements

The portfolio consists of short assessments and exercises completed individually that demonstrate your understanding and skills in key topics from the module. These assessments are structured as follows:

1. **Bayesian programming for perception:** You are expected to demonstrate knowledge of how uncertainty is dealt with using Bayesian modelling by exploring in practice a given perceptual example.
2. **Navigation with ROS2 in Gazebo simulation:** You are expected to demonstrate the ability to implement a well-defined navigation and collision avoidance strategy.
3. **Mapping:** You are expected to explore off-the-shelf 3D mapping frameworks which would eventually be used for SLAM in a simulated Gazebo environment using ROS2.

The portfolio submission will consist of a report with a section/chapter for each component and a video presentation showcasing the respective outcomes.

Failing to submit the presentation will result in not meeting the passing criteria for the assessment.

The following deliverables are mandatory:

- 1 – **Individual report** to be submitted to the NOW Dropbox **Draft Report folder** to detect and reduce similarity of your presentation. This must be either a PDF or a MS Word document.
- 2 – **Individual report** to be submitted to the NOW Dropbox **Main Report folder**. This is your official submission, and you **should not have a high similarity match with other student presentations**. This must be either a PDF or a MS Word Document.
- 3 – A compressed file containing the projects to be submitted to the NOW Dropbox **Source Code folder**.

4 – **A video presentation narrated by the student explaining the outcomes of each part of the Main Report** to be submitted to the NOW Dropbox.

Note that:

- The code **MUST** also be stored in git repository on <https://olympus.ntu.ac.uk>. Please refer to the labs to learn how to setup your git repository. This will enable you to recover quickly should the system fail and allow you to backtrack if your development goes astray.
- The source must compile and run in the module docker containers (**pedrombmachado/lubuntu_ros:comp30271** or **pedrombmachado/lubuntu_ros:arm64_comp30271**)
- Keep evidence of the submission of your assignment, and a copy of your assignment in case of the unlikely event of any loss.
- **Code will only be analysed if the report and/or demo presentation do not clearly and consistently demonstrate reported functionalities working as claimed.**

You are expected to submit the report individually, and, for each component, it should include descriptions of the theoretical background involved, your approach to solving the task, any system/code design details, and results (i.e. generated outputs, performance evaluation, etc.).

The report must include a table of contents and each section must start on a new page and include a respective reference list. The report must be no more than 45 pages long (i.e., not including the cover page and table of contents, which means an average of 15 pages for each section). **It must not contain any code snippets**. To describe code, use code design illustration languages such as UML.

II. Assessment Criteria

Component	Sub-Component	Exceptional First	First	2.1	2.2	3 rd Pass	Fail	Zero
Portfolio 1	Implementation	Able to effectively apply Bayesian programming principles to solve perceptual problems.	Shows an effective application of Bayesian programming principles to solve perceptual problems.	Demonstrates a competent application of Bayesian programming principles to solve perceptual problems.	Demonstrates a basic application of Bayesian programming principles to solve perceptual problems.	Demonstrates a rudimentary application of Bayesian programming principles with perceptual problems.	Work of low merit or demo did not show enough functionality.	Work of no merit or absent; work not demonstrated; penalty in some misconduct cases.
	Reporting & Presentation	Demonstrates a deep understanding of uncertainty handling using Bayesian models.	Demonstrates a good understanding of uncertainty handling using Bayesian models.	Adequate understanding of uncertainty handling using Bayesian models.	Limited understanding of uncertainty handling using Bayesian models.	Little understanding of uncertainty handling using Bayesian models.	Work of low merit; failed to show even a basic understanding of the required knowledge.	Work of no merit or absent; work not submitted; penalty in some misconduct cases.
Portfolio 2	Implementation	Able to implement a comprehensive navigation and collision avoidance strategy in a simulated environment.	Shows proficient use of ROS2 and Gazebo for navigation and collision avoidance.	Adequate use of ROS2 and Gazebo for navigation and collision avoidance.	Basic use of ROS2 and Gazebo for navigation and collision avoidance.	Limited use of ROS2 and Gazebo; either navigation or collision avoidance not working as required.	Work of low merit or demo did not show enough functionality.	Work of no merit or absent; work not demonstrated; penalty in some misconduct cases.

	<i>Reporting & Presentation</i>	Demonstrates thorough understanding and implementation of ROS2 and Gazebo.	Demonstrates a solid grasp of navigation and collision avoidance strategy in a simulated environment.	Shows a competent understanding of navigation and collision avoidance strategy in a simulated environment.	Shows a satisfactory understanding of navigation and collision avoidance strategy in a simulated environment.	Shows a basic understanding of navigation and collision avoidance strategy in a simulated environment.	Work of low merit; failed to show even a basic understanding of the required knowledge.	Work of no merit or absent; work not submitted; penalty in some misconduct cases.
Portfolio 3	<i>Implementation</i>	Successfully explores off-the-shelf mapping frameworks in a simulated Gazebo environment using ROS2.	Effectively explores off-the-shelf mapping frameworks in a simulated Gazebo environment using ROS2.	Adequately explores off-the-shelf mapping frameworks in a simulated Gazebo environment using ROS2.	Explores off-the-shelf mapping frameworks in a simulated Gazebo environment using ROS2 to a basic extent.	Explores off-the-shelf mapping frameworks in a simulated Gazebo environment using ROS2 to a limited extent.	Work of low merit or demo did not show enough functionality.	Work of no merit or absent; work not demonstrated; penalty in some misconduct cases.
	<i>Reporting & Presentation</i>	Demonstrates an advanced understanding of mapping concepts and frameworks.	Demonstrates a good understanding of mapping concepts and frameworks.	Demonstrates a competent understanding of mapping concepts and frameworks.	Demonstrates a basic understanding of mapping concepts and frameworks.	Demonstrates a limited understanding of mapping concepts and frameworks.	Work of low merit; failed to show even a basic understanding of the required knowledge.	Work of no merit or absent; work not submitted; penalty in some misconduct cases.

III. Feedback Opportunities

Formative (Whilst you're working on the coursework)

- Attend lab sessions and ask questions about your implementation.
- Sign up for tutor meetings to get specific feedback as you work on assessments.
- Use peer discussions and code reviews to improve your solutions.
- Ask questions and discuss implementation issues on NOW discussion forums.

Summative (After you've submitted the coursework)

- Written feedback will be provided through NOW within 10 working days.
- Feedback will relate to the assessment criteria and highlight areas done well and those needing improvement.

IV. Resources that may be useful

Referencing styles please use Harvard as detailed [here](#)

Guide to planning your time [here](#) and an automated planner [here](#)

Guidance on avoiding cheating is [here](#)

Remember to use Outlook or physical calendars to block out time between lectures and labs to work on this coursework.

V. Moderation

The Moderation Process

All assessments are subject to a two-stage moderation process. Firstly, any details related to the assessment (e.g., clarity of information and the assessment criteria) are considered by an independent person (usually a member of the module team). Secondly, the grades awarded are considered by the module team to check for consistency and fairness across the cohort for the piece of work submitted.

VI. Aspects for Professional Development

Successfully completing this coursework will generally cover the learning outcomes expected for the module. It will specifically provide you with the following knowledge and skills:

- Entry-level grasp of the fundamentals in CC & ACS development and design, namely edge-fog-cloud computing, machine learning, probabilistic methods and artificial neural networks (deep learning).
- The capacity of being able to independently research topics relating to CC & ACS.
- In-depth skills in system architecture design, mathematical modelling and programming required to implement CC & ACS.
- The capacity of taking in and integrating know-how both from computer science and engineering and from human sciences and nature.
- The capability of testing applications targeting real robots in simulation environments.
- The ability of working effectively within a team.

Many of these are useful transferable skills for employment applications.